Tanishq Dahiya

AI Engineer · Agents & Tool-Calling · RAG · LLM Evaluation & Safety · Python
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SUMMARY

AI engineer who ships to production rather than to notebooks. Built and deployed Kaia, a live multi-channel assistant handling text, browser voice and real phone calls, with Claude tool-calling across Google Workspace and model-written pandas running in a resource-capped sandbox. Designed an adversarial LLM safety benchmark that autonomously scored 8,225 responses across 5 frontier models, exposing a 60.2% intent-level inconsistency rate. MSc Computing (AI), First Class Honours, Dublin City University.

EDUCATION

MSc Computing (AI) | First Class Honours (1:1) 2025 – 2026
Dublin City University, Ireland

BSc Computer Science | First Division 2021 – 2024
University of Delhi, India

Additional: Amazon AI/ML Summer School 2023

RESEARCH

LLM Safety & Multi-Agent Evaluation (ISCE) | MSc Practicum, DCU 2025 – 2026

- Safety benchmarks test one phrasing at a time, so a model that refuses one wording but complies with a semantically equivalent rewrite passes undetected. Designed 329 adversarial seed prompts grounded in 14 MITRE ATT&CK techniques, expanded to 1,645 variants via 4 controlled paraphrasing strategies (lexical, logic reversal, syntactic, contextual).
- Built a multi-agent evaluation pipeline: 5 autonomous agents each calling a different production LLM API (Claude, GPT-4o, DeepSeek, Mistral, Qwen) with independent rate limiting, structured JSON parsing and fault-tolerant recovery — 8,225 requests in parallel with no human intervention — scored by a 3-judge G-EVAL consensus system producing 24,675 verdicts.
- Proposed the IIR (Intent-level Inconsistency Rate) metric: 60.2% strict inconsistency, with the safest models (MSA 59%) the least consistent (IIR 72.6%) — evidence that safety alignment and behavioural consistency are independent properties. Authored the thesis with full reproducibility documentation.

PROJECTS

Kaia — Production AI Assistant (live) *FastAPI · Claude · Next.js · Deepgram · Telnyx · Supabase · Docker · Hetzner*

- Deployed a multi-channel AI assistant — text chat, browser voice and real inbound/outbound phone calls — to a Hetzner VPS behind Caddy: 16 tools across Google Calendar, Gmail and Sheets CRM, Postgres with row-level security on every table, and per-provider cost tracking. ~10k LOC across FastAPI and Next.js 15.
- Routed spreadsheet questions off the RAG path after retrieval proved unsound for aggregation: the model inspects the real schema, writes pandas, and the code runs in an isolated subprocess as an unprivileged user under CPU, memory, file-descriptor and process limits — totals computed against the file, not recalled from chunks.
- Cut voice latency to 0.20s first-audio by replacing per-sentence HTTP speech calls with a single persistent WebSocket fed by streaming LLM tokens; made it safe for external users with revocable per-person access codes, a restricted guest tool allowlist, Fernet-encrypted OAuth tokens and a daily spend ceiling enforced in code.

Hybrid RAG over EU & Irish Law, with Automated Evaluation *BM25 (SQLite FTS5) · FAISS · Reciprocal Rank Fusion · Evaluation*

- Legal questions come in two incompatible shapes: exact citations ("AI Act Article 6(2)", "Stamp 1G") and conceptual queries sharing no vocabulary with the statute that answers them. Fused BM25 and dense retrieval with Reciprocal Rank Fusion, which uses ranks rather than scores so two incompatible scales never need normalising — recall@10 93.3% against 91.7% and 86.7% for the single retrievers, multi-hop recall 80% → 100%.
- Built the evaluation layer as evidence rather than decoration: 68 questions including 10 unanswerable traps measuring abstention; 95.2% faithfulness, 4.8% hallucination, 100% of citations resolving to a real source. Used it to delete one of the project's own features — a grounding-score retry gate that measurement showed never fired, and was inversely correlated with the quality it gated on.

Prompt Firewall — Real-Time Prompt Injection Detection *FastAPI · Layered Classification · Docker · CI*

- Two-layer filter sitting between users and an LLM: a sub-millisecond regex pass catching known injection signatures, base64 blobs and Unicode homoglyphs, falling through to an LLM classifier returning a structured verdict, category and confidence — the cheap layer handles the common case, the expensive one only sees ambiguity. Shipped with a test suite and CI.

Also: Multi-Agent Research Assistant (LangGraph · ChromaDB · Groq) — three agents with structured handoffs, each validating the last, behind a human approval gate. AI-Generated Image Detection (EfficientNet-B3 · CLIP ViT-B/16 · XGBoost) — 3-stream ensemble, macro-F1 0.8751 on 4,800 images, +5.4pp over the best single model. Analytics Pipeline (dbt · Snowflake) — 15+ dbt models on a star schema, 20+ automated quality tests, CI/CD.

TECHNICAL SKILLS

Agentic AI & LLMs: Claude/OpenAI tool-calling, agent orchestration (LangGraph, CrewAI), RAG (LangChain, LlamaIndex), hybrid retrieval (BM25 + dense, RRF), vector stores (FAISS, ChromaDB), LoRA fine-tuning, prompt engineering, guardrails, human-in-the-loop
Evaluation & Quality: Benchmark design, adversarial testing, G-EVAL consensus judging, faithfulness / hallucination / abstention metrics, safety and fairness assessment, model monitoring (IIR, F1, nDCG)
Backend & Real-Time: Python, FastAPI, REST, WebSockets, streaming pipelines, async and concurrency, Postgres (Supabase, RLS), SQLite FTS5, subprocess sandboxing and resource limits, OAuth 2.0, webhooks
Languages & ML: Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow), SQL, TypeScript, Bash, R
Cloud, Data & DevOps: Docker Compose, Hetzner VPS, Caddy/Nginx, AWS (S3, EC2), Azure, GitHub Actions CI/CD, Linux, dbt, Snowflake, Spark/PySpark, Databricks, Airflow, MLflow (exposure)
Responsible AI & Tooling: PII handling, content safety, prompt-injection defence, least-privilege tool design, spend controls; Claude Code, Copilot, Cursor

Graduated Sep 2026 | Eligible to work full-time in Ireland (Stamp 1G) | Comfortable relocating | Full clean Irish driver's licence